

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

REC_2028_OOPS using Java_Week 10_CY

Attempt : 1
Total Mark : 40
Marks Obtained : 40

Section 1 : COD

1. Problem Statement

Arjun is working on a program that checks if one set of numbers is a subset of another. If Set B is a subset of Set A, the program should print "YES" followed by the sorted elements of Set B. If Set B is not a subset of Set A, the program should print "NO" followed by the average of all elements from both sets combined, rounded to two decimal places.

Implement a class Solution with the required method to perform the subset check using TreeSet in Java.

Input Format

The first line contains an integer n - the number of elements in Set A.

The second line contains n space-separated integers - the elements of Set A.

The third line contains an integer m - the number of elements in Set B.

The fourth line contains m space-separated integers - the elements of Set B.

Output Format

If Set B is a subset of Set A, print "YES" followed by the sorted values of Set B.

Otherwise, print "NO" followed by the average of all numbers in both sets (rounded to two decimal places).

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

1 2 3 4 5

3

2 3 5

Output: YES 2 3 5

Answer

```
import java.util.*;
```

```
class Solution {
    public static void checkSubset(TreeSet<Integer> setA, TreeSet<Integer> setB,
int totalElements, double sum) {
        if (setA.containsAll(setB)) {
            System.out.print("YES ");
            for (int num : setB) {
                System.out.print(num + " ");
            }
            System.out.println();
        } else {
            double average = sum / totalElements;
            System.out.printf("NO %.2f%n", average);
        }
    }
}
```

```
class Main {
```

```

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int n = sc.nextInt();
    TreeSet<Integer> setA = new TreeSet<>();
    long sum = 0;
    for (int i = 0; i < n; i++) {
        int num = sc.nextInt();
        setA.add(num);
        sum += num;
    }
    int m = sc.nextInt();
    TreeSet<Integer> setB = new TreeSet<>();
    for (int i = 0; i < m; i++) {
        int num = sc.nextInt();
        setB.add(num);
        sum += num;
    }
    Solution.checkSubset(setA, setB, n + m, sum);
    sc.close();
}
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Tony is an e-learning platform administrator, he oversees the user ratings for various online courses offered in the platform.

To enhance user experience, you should assist him in utilizing a HashMap to store course ratings given by learners. Regularly, he analyzes this data to identify the highest and lowest-rated courses, enabling targeted improvements and ensuring the quality of the educational content. This process assists in maintaining a competitive and engaging online learning environment for the users.

Input Format

The input consists of a string representing the course name followed by a double value representing the course's rating, in separate lines.

The input is terminated by entering "done".

Output Format

The first line of output prints the string "Highest Rated Course: " followed by the highest-rated course.

The second line prints the string "Lowest Rated Course: " followed by the lowest-rated courses.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: DSA

4.0

OOPS

4.2

C

3.2

done

Output: Highest Rated Course: OOPS

Lowest Rated Course: C

Answer

```
import java.util.HashMap;
```

```
import java.util.Map;
```

```
import java.util.Scanner;
```

```
class CourseAnalyzer {
```

```
    public Map<String, String>
```

```
    identifyHighestAndLowestRatedCourses(Map<String, Double> courseRatings) {
```

```
        double highestRating = Double.MIN_VALUE;
```

```
        double lowestRating = Double.MAX_VALUE;
```

```
        String highestRatedCourse = "";
```

```
        String lowestRatedCourse = "";
```

```
        for (Map.Entry<String, Double> entry : courseRatings.entrySet()) {
```

```
            String course = entry.getKey();
```

```
            double rating = entry.getValue();
```

```

        if (rating > highestRating) {
            highestRating = rating;
            highestRatedCourse = course;
        }
        if (rating < lowestRating) {
            lowestRating = rating;
            lowestRatedCourse = course;
        }
    }
}

Map<String, String> result = new HashMap<>();
result.put("highest", highestRatedCourse);
result.put("lowest", lowestRatedCourse);
return result;
}
}

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        Map<String, Double> courseRatings = new HashMap<>();

        while (true) {
            String courseName = scanner.nextLine();
            if (courseName.equalsIgnoreCase("done")) {
                break;
            }
            double rating = Double.parseDouble(scanner.nextLine().trim());
            courseRatings.put(courseName, rating);
        }

        CourseAnalyzer analyzer = new CourseAnalyzer();
        Map<String, String> result =
analyzer.identifyHighestAndLowestRatedCourses(courseRatings);

        System.out.printf("Highest Rated Course: %s\n", result.get("highest"));
        System.out.printf("Lowest Rated Course: %s", result.get("lowest"));

        scanner.close();
    }
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

A linguist named Meera is classifying a list of words based on their first character. She wants to store words grouped by their starting letter using a TreeMap so that the groups appear in sorted order of characters (i.e., 'a' to 'z'). For each letter, all words starting with that letter should be stored in the order they appear.

Implement the logic inside a class named WordClassifier using the TreeMap<Character, List<String>> collection.

Input Format

The first line of the input contains an integer n, representing the number of words.

The next n lines each contain a word.

Output Format

The first line of the output prints: "Grouped Words by Starting Letter:"

The next lines print each character key and its list of words in the format:

"letter: word1 word2 word3..."

..."

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

dog

deer

cat

cow

camel

Output: Grouped Words by Starting Letter:

c: cat cow camel

d: dog deer

Answer

```
import java.util.*;

class WordClassifier {
    public void classifyWords(List<String> words) {
        TreeMap<Character, List<String>> map = new TreeMap<>();

        for (String word : words) {
            char initial = word.charAt(0);
            if (!map.containsKey(initial)) {
                map.put(initial, new ArrayList<>());
            }
            map.get(initial).add(word);
        }

        System.out.println("Grouped Words by Starting Letter:");
        for (Map.Entry<Character, List<String>> entry : map.entrySet()) {
            System.out.print(entry.getKey() + ": ");
            for (String word : entry.getValue()) {
                System.out.print(word + " ");
            }
            System.out.println();
        }
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = Integer.parseInt(sc.nextLine());

        List<String> words = new ArrayList<>();
        for (int i = 0; i < n; i++) {
            words.add(sc.nextLine());
        }

        WordClassifier classifier = new WordClassifier();
        classifier.classifyWords(words);
    }
}
```

Status : Correct

Marks : 10/10

4. Problem Statement

David is managing an employee database where each employee has a unique ID, name, and department. He wants to ensure that duplicate employee IDs are not added to the system. Implement a Java program that allows adding employees to the system, displaying all employees, and checking if an employee exists based on the given ID.

Implement a class `EmployeeDatabase` that contains a `HashSet` to store employee records. The `Employee` class should be a user-defined object containing employee details. The main class should handle user operations and interact with the `EmployeeDatabase` class.

Input Format

The first line contains an integer `n` representing the number of employees to be added.

The next `n` lines follow, each containing:

1. An integer `employee_id`
2. A string `name`
3. A string `department`

The next line contains an integer `m` representing the number of queries.

The next `m` lines follow, each containing an employee ID to check for existence.

Output Format

The output prints a list of all employees added in the format:

"ID: <employee_id>, Name: <name>, Department: <department>"

For each query, output "Employee exists" if the ID is found, otherwise "Employee not found".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 3

101 John IT

102 Alice HR

103 Bob Finance

2

101

104

Output: ID: 101, Name: John, Department: IT

ID: 102, Name: Alice, Department: HR

ID: 103, Name: Bob, Department: Finance

Employee exists

Employee not found

Answer

```
import java.util.*;
```

```
class Employee {
```

```
    int employeeId;
```

```
    String name, department;
```

```
    public Employee(int employeeId, String name, String department) {
```

```
        this.employeeId = employeeId;
```

```
        this.name = name;
```

```
        this.department = department;
```

```
    }
```

```
    public int hashCode() {
```

```
        return Objects.hash(employeeId);
```

```
    }
```

```
    public boolean equals(Object obj) {
```

```
        if (this == obj) return true;
```

```
        if (obj == null || getClass() != obj.getClass()) return false;
```

```
        Employee e = (Employee) obj;
```

```
        return this.employeeId == e.employeeId;
```

```
    }
```

```
    public String toString() {
```

```

        return "ID: " + employeeId + ", Name: " + name + ", Department: " +
department;
    }
}

```

```

class EmployeeDatabase {
    HashSet<Employee> employees = new HashSet<>();

    public void addEmployee(int id, String name, String department) {
        employees.add(new Employee(id, name, department));
    }

    public void displayEmployees() {
        for (Employee e : employees) {
            System.out.println(e);
        }
    }

    public boolean checkEmployee(int id) {
        return employees.contains(new Employee(id, "", ""));
    }
}

```

```

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        EmployeeDatabase db = new EmployeeDatabase();
        int n = sc.nextInt();
        for (int i = 0; i < n; i++) {
            int id = sc.nextInt();
            String name = sc.next();
            String department = sc.next();
            db.addEmployee(id, name, department);
        }
        db.displayEmployees();
        int m = sc.nextInt();
        for (int i = 0; i < m; i++) {
            int id = sc.nextInt();
            if (db.checkEmployee(id))
                System.out.println("Employee exists");
            else
                System.out.println("Employee not found");
        }
    }
}

```

```
        sc.close();  
    }  
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

REC_2028_OOPS using Java_Week 10_MCQ

Attempt : 1
Total Mark : 15
Marks Obtained : 15

Section 1 : MCQ

1. Which method retrieves the lowest key in a TreeMap?

Answer

firstKey()

Status : Correct

Marks : 1/1

2. Which of the following is true about TreeMap?

Answer

It maintains natural ordering

Status : Correct

Marks : 1/1

3. Which of the following allows null keys in Java?

Answer

HashMap

Status : Correct

Marks : 1/1

4. Which of the following is true about HashMap?

Answer

It is not synchronized

Status : Correct

Marks : 1/1

5. What will be the output of the following code?

```
import java.util.*;
class Main {
    public static void main(String[] args) {
        HashMap<String, String> map = new HashMap<>();
        map.put("A", "Apple");
        map.put("B", "Banana");
        map.put("C", "Cherry");
        map.replace("B", "Blueberry");
        System.out.println(map);
    }
}
```

Answer

{A=Apple, B=Blueberry, C=Cherry}

Status : Correct

Marks : 1/1

6. Which method removes all elements from a Set?

Answer

clear()

Status : Correct

Marks : 1/1

7. What happens if two keys have the same hash code in a HashMap?

Answer

A linked list is used to store values with the same hash

Status : Correct

Marks : 1/1

8. What will happen if you add elements in descending order in a TreeSet?

Answer

They are sorted in ascending order

Status : Correct

Marks : 1/1

9. What happens when you add duplicate elements to a HashSet?

Answer

The duplicate is ignored

Status : Correct

Marks : 1/1

10. Which statement is true about HashSet and TreeSet?

Answer

TreeSet provides sorted elements

Status : Correct

Marks : 1/1

11. How does HashSet check for duplicate elements?

Answer

Using equals() and hashCode()

Status : Correct

Marks : 1/1

12. What will be the output of the following code?

```
import java.util.*;
class Main {
    public static void main(String[] args) {
        HashMap<String, Integer> map = new HashMap<>();
        map.put("X", 10);
        map.put("Y", 20);
        map.put("Z", 30);
        map.remove("Y");
        System.out.println(map);
    }
}
```

Answer

{X=10, Z=30}

Status : Correct

Marks : 1/1

13. What is the time complexity of retrieving an element from a HashSet?

Answer

O(1)

Status : Correct

Marks : 1/1

14. What will happen if you add a null element to a TreeSet?

Answer

An exception occurs

Status : Correct

Marks : 1/1

15. What will be the output of the following code?

```
import java.util.*;
class Main {
    public static void main(String[] args) {
```

```
HashMap<String, Integer> map = new HashMap<>();  
map.put("A", 1);  
map.put("B", 2);  
map.put("C", 3);  
System.out.println(map.containsKey("B"));  
}  
}
```

Answer

true

Status : Correct

Marks : 1/1

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2024_28_III_OOPS Using Java Lab

REC_2028_OOPS using Java_Week 10_PAH

Attempt : 1
Total Mark : 30
Marks Obtained : 30

Section 1 : Coding

1. Problem Statement

Riya is building a calendar event scheduler where each event is stored in chronological order using a TreeMap. The key represents the event time in 24-hour format (HH:MM), and the value is the event description.

She wants the system to:

Automatically sort events by time. Avoid duplicate time entries — if a duplicate time is entered, ignore the new entry. Print all scheduled events in order.

Implement this logic using a class named EventManager.

Input Format

The first line of the input contains an integer n , representing the number of events.

The next n lines each contain a string in the format: "HH:MM Description"

(Example: 09:00 TeamMeeting).

Output Format

The first line of the output prints "Scheduled Events:"

The next k lines print each event in the format: "HH:MM - Description"

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

09:00 TeamMeeting

13:30 LunchBreak

11:00 ProjectUpdate

09:00 Standup

15:00 ClientCall

Output: Scheduled Events:

09:00 - TeamMeeting

11:00 - ProjectUpdate

13:30 - LunchBreak

15:00 - ClientCall

Answer

```
import java.util.*;
class EventManager {
    TreeMap<String, String> schedule;

    public EventManager() {
        schedule = new TreeMap<>();
    }

    public void addEvent(String time, String description) {
        if (!schedule.containsKey(time)) {
            schedule.put(time, description);
        }
    }
}
```

```

    public void printSchedule() {
        System.out.println("Scheduled Events:");
        for (Map.Entry<String, String> entry : schedule.entrySet()) {
            System.out.println(entry.getKey() + " - " + entry.getValue());
        }
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = Integer.parseInt(sc.nextLine());

        EventManager manager = new EventManager();

        for (int i = 0; i < n; i++) {
            String line = sc.nextLine();
            int spaceIndex = line.indexOf(' ');
            String time = line.substring(0, spaceIndex);
            String desc = line.substring(spaceIndex + 1);
            manager.addEvent(time, desc);
        }

        manager.printSchedule();
    }
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Sarah is working on a spam detection system that analyzes incoming messages for unique patterns. Spammers often use repetitive character sequences, making it important to identify the first non-repeating character in a message.

Given a string, Sarah needs to determine the first character that appears only once. If all characters repeat, the system should return -1.

She decides to use a HashMap to efficiently track character frequencies

and find the solution.

Input Format

The first line contains an integer N representing , the length of the string.

The second line contains a string of N lowercase English letters (a-z).

Output Format

The output prints a character representing the first non-repeating character. If none exist, print -1.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 10
abacabadac

Output: d

Answer

```
import java.util.*;
class NonRepeatingCharacterFinder {

    public char findFirstNonRepeatingCharacter(String str) {
        HashMap<Character, Integer> charCount = new HashMap<>();

        for (char ch : str.toCharArray()) {
            charCount.put(ch, charCount.getOrDefault(ch, 0) + 1);
        }

        for (char ch : str.toCharArray()) {
            if (charCount.get(ch) == 1) {
                return ch;
            }
        }

        return '\0';
    }
}
```

```

class FirstNonRepeatingCharacter {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int N = sc.nextInt();
        String str = sc.next();

        NonRepeatingCharacterFinder finder = new NonRepeatingCharacterFinder();
        char result = finder.findFirstNonRepeatingCharacter(str);

        if (result == '\0') {
            System.out.println(-1);
        } else {
            System.out.println(result);
        }

        sc.close();
    }
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

A university maintains a list of student records and wants to store them in a sorted manner based on their GPA. If two students have the same GPA, they should be further sorted by their name in lexicographical order. Implement a program that uses a TreeSet to store student records and ensures unique student IDs.

Input Format

The first line contains an integer N - the number of students.

The next N lines contain details of each student in the format: "StudentID Name GPA"

- StudentID (Integer) - A unique identifier.
- Name (String) - The student's name (can contain spaces).
- GPA (Double) - The Grade Point Average.

Output Format

The output prints the list of students in ascending order of GPA.

If two students have the same GPA, sort them by name.

Print details in the format: "StudentID Name GPA" in the output, GPA is rounded to two decimal places.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

101 John 8.5

102 Alice 9.1

103 Bob 8.5

104 Zoe 7.3

105 Charlie 9.1

Output: 104 Zoe 7.30

103 Bob 8.50

101 John 8.50

102 Alice 9.10

105 Charlie 9.10

Answer

```
import java.util.*;
class Student implements Comparable<Student> {
    int studentID;
    String name;
    double gpa;

    public Student(int studentID, String name, double gpa) {
        this.studentID = studentID;
        this.name = name;
        this.gpa = gpa;
    }

    public int compareTo(Student other) {
        if (this.gpa != other.gpa) {
```

```

        return Double.compare(this.gpa, other.gpa);
    }
    return this.name.compareTo(other.name);
}

public String toString() {
    return studentID + " " + name + " " + String.format("%.2f", gpa);
}
}

class UniversityRecords {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        sc.nextLine();
        TreeSet<Student> studentSet = new TreeSet<>();
        for (int i = 0; i < n; i++) {
            int id = sc.nextInt();
            String name = sc.next();
            double gpa = sc.nextDouble();
            studentSet.add(new Student(id, name, gpa));
        }
        for (Student s : studentSet) {
            System.out.println(s);
        }
        sc.close();
    }
}

```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 10_Q1

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : COD

1. Problem Statement

A city traffic management system needs to track vehicles entering a toll booth. Each vehicle is uniquely identified by its registration number. The system should allow adding vehicles to a record, ensuring that no duplicate registration numbers exist. The vehicles should be stored in a HashSet, which does not guarantee any specific order.

Your task is to implement a program using a HashSet that allows adding vehicle details and displaying the records.

Input Format

The first line of input contains an integer N - the number of vehicles.

The next N lines contain details of each vehicle in the format: "RegNumber

OwnerName VehicleType"

1. RegNumber (String) - A unique registration number (Alphanumeric).
2. OwnerName (String) - The name of the vehicle owner.
3. VehicleType (String, Car, Bike, or Truck) - The type of vehicle.

If a vehicle with the same registration number is already present, ignore the duplicate entry.

Output Format

The output prints the unique vehicle records in any order (since HashSet does not maintain order).

Output format: "RegNumber OwnerName VehicleType"

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

KA01AB1234 John Car

MH02CD5678 Alice Bike

DL03EF9012 Bob Truck

TN04GH3456 Mike Car

KA01AB1234 John Car

Output: TN04GH3456 Mike Car

KA01AB1234 John Car

MH02CD5678 Alice Bike

DL03EF9012 Bob Truck

Answer

```
import java.util.*;
class Vehicle {
    String regNumber;
    String ownerName;
    String vehicleType;
    private static HashSet<Vehicle> vehicleSet = new HashSet<>();
    public Vehicle(String regNumber, String ownerName, String vehicleType) {
        this.regNumber = regNumber;
        this.ownerName = ownerName;
```

```

        this.vehicleType = vehicleType;
    }
    public static void addVehicle(String regNumber, String ownerName, String
vehicleType) {
        vehicleSet.add(new Vehicle(regNumber, ownerName, vehicleType));
    }
    public static void displayVehicles() {
        for (Vehicle v : vehicleSet) {
            System.out.println(v);
        }
    }
    public boolean equals(Object obj) {
        if (this == obj) return true;
        if (obj == null || getClass() != obj.getClass()) return false;
        Vehicle vehicle = (Vehicle) obj;
        return regNumber.equals(vehicle.regNumber);
    }
    public int hashCode() {
        return Objects.hash(regNumber);
    }
    public String toString() {
        return regNumber + " " + ownerName + " " + vehicleType;
    }
}class TollBoothSystem {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        sc.nextLine();
        for (int i = 0; i < n; i++) {
            String regNumber = sc.next();
            String ownerName = sc.next();
            String vehicleType = sc.next();
            Vehicle.addVehicle(regNumber, ownerName, vehicleType);
        }
        Vehicle.displayVehicles();
        sc.close();
    }
}

```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 10_Q2

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : COD

1. Problem Statement

John is organizing a fruit festival, and the quantities of various fruits are stored in a HashMap where fruit names are keys and quantities are values.

Help him develop a program to find the total quantity of fruits for the festival by summing up the values in the HashMap.

Input Format

The input consists of fruit quantities in the format 'fruitName:quantity', where fruitName is the name of the fruit(a string), and quantity is a double value representing the quantity.

The input is terminated by entering "done".

Output Format

The output prints a double value, representing the sum of values in the HashMap, rounded off to two decimal places.

If the value is not numeric, print "Invalid input".

If any special characters other than ':' are entered, print "Invalid format".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: Banana:15.2

Orange:56.3

Mango:47.3

done

Output: 118.80

Answer

```
import java.util.Scanner;
import java.util.Map;
import java.util.HashMap;
class ValueProcessor {
    public static Map<String, Double> readValues(Scanner scanner) {
        Map<String, Double> valueMap = new HashMap<>();
        while (true) {
            String input = scanner.nextLine();
            if (input.toLowerCase().equals("done")) {
                break;
            }
            String[] pair = input.split(":");
            if (pair.length == 2) {
                String key = pair[0].trim();
                try {
                    double value = Double.parseDouble(pair[1].trim());
                    valueMap.put(key, value);
                } catch (NumberFormatException e) {
                    System.out.println("Invalid input");
                    return null;
                }
            } else {
```

```

        System.out.println("Invalid format");
        return null;
    }
}
return valueMap;
}

public static double calculateSum(Map<String, Double> valueMap) {
    double sum = 0;
    for (double value : valueMap.values()) {
        sum += value;
    }
    return sum;
}
}

class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        Map<String, Double> valueMap = ValueProcessor.readValues(scanner);
        if (valueMap != null) {
            double sum = ValueProcessor.calculateSum(valueMap);
            System.out.printf("%.2f\n", sum);
        }
        scanner.close();
    }
}

```

Status : Correct

Marks : 10/10

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Scan to verify results



2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 10_Q3

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : COD

1. Problem Statement

Priya is analyzing encrypted messages in a research project. She wants to analyze the frequency of each character in a given paragraph. The characters should be stored in a TreeMap so that the output is sorted in ascending order of characters automatically.

You are required to build a Java program that:

Uses a `TreeMap<Character, Integer>` to count how many times each character appears in the message. Ignores spaces and considers only alphabets (case-sensitive). Outputs the frequencies of characters in sorted order.

You must use a TreeMap in the class named MessageAnalyzer.

Input Format

The first line of input contains an integer n, the number of lines in the message.

The next n lines each contain a string (the encrypted message line).

Output Format

The first line of output prints: "Character Frequency:"

Then print each character and its frequency in the format: "<character>: <count>"

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 2

Hello World

Java

Output: Character Frequency:

H: 1

J: 1

W: 1

a: 2

d: 1

e: 1

l: 3

o: 2

r: 1

v: 1

Answer

```
import java.util.*;
class MessageAnalyzer {
    public void analyzeMessageFrequency(List<String> lines) {
        TreeMap<Character, Integer> frequencyMap = new TreeMap<>();

        for (String line : lines) {
            for (char ch : line.toCharArray()) {
                if (Character.isLetter(ch)) {
                    frequencyMap.put(ch, frequencyMap.getDefault(ch, 0) + 1);
                }
            }
        }
    }
}
```

```

    }

    System.out.println("Character Frequency:");
    for (Map.Entry<Character, Integer> entry : frequencyMap.entrySet()) {
        System.out.println(entry.getKey() + ": " + entry.getValue());
    }
}

}

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = Integer.parseInt(sc.nextLine());
        List<String> lines = new ArrayList<>();
        for (int i = 0; i < n; i++) {
            lines.add(sc.nextLine());
        }

        MessageAnalyzer analyzer = new MessageAnalyzer();
        analyzer.analyzeMessageFrequency(lines);
    }
}

```

Status : Correct

Marks : 10/10

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Scan to verify results



2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 10_Q4

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : COD

1. Problem Statement

In a ticket reservation system, you store the available seat numbers in a TreeSet. Users input their desired seat number, and the program checks whether the chosen seat is available.

Using a TreeSet ensures quick and efficient verification of seat availability, ensuring a smooth and organized ticket booking process.

Input Format

The first line of input contains a single integer n , representing the number of available seats.

The second line contains n space-separated integers, representing the available seat numbers.

The third line contains an integer *m*, representing the seat number that needs to be searched.

Output Format

The output displays "[*m*] is present!" if the given seat is available. Otherwise, it displays "[*m*] is not present!"

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 4

2 4 5 6

5

Output: 5 is present!

Answer

```
import java.util.Set;
import java.util.TreeSet;
import java.util.Scanner;
class NumberChecker {
    private Set<Integer> numberSet;

    public NumberChecker(Set<Integer> numberSet) {
        this.numberSet = numberSet;
    }

    public void addNumbers(int[] numbers) {
        for (int number : numbers) {
            numberSet.add(number);
        }
    }

    public String checkNumber(int number) {
        return numberSet.contains(number) ? number + " is present!" : number + " is not present!";
    }
}
class Main {
    public static void main(String[] args) {
```

```
Scanner scanner = new Scanner(System.in);
int numberOfElements = scanner.nextInt();
int[] numbers = new int[numberOfElements];

for (int i = 0; i < numberOfElements; i++) {
    numbers[i] = scanner.nextInt();
}

int elementToCheck = scanner.nextInt();
scanner.close();

Set<Integer> numberSet = new TreeSet<>();
NumberChecker numberChecker = new NumberChecker(numberSet);
numberChecker.addNumbers(numbers);

System.out.println(numberChecker.checkNumber(elementToCheck));
}
```

Status : Correct

Marks : 10/10